

Zoo Gaming Mechanics

Tamagotchi-Style Pet Care Meets Game-Fi

Antje Worrying
Founder, Zoo Labs Foundation
antje@zoo.ngo

October 31, 2021

Abstract

This element paper expands the gaming mechanics from the Zoo Whitepaper [1]. Zoo’s game-play weaves Tamagotchi-style pet care — feed, grow, breed — with collateralization, daily yield, and a generation-limited breeding curve. We document the lifecycle, the egg drop mechanics, the boost ladder, and the play-and-earn games that anchor user engagement.

1 The lifecycle

A Zoo animal proceeds through four stages:

1. **Egg.** Acquired in a drop. Three rarity tiers: Sublime (1 ETH, 5%), Rare (0.5 ETH, 15%), Endangered (0.05–0.1 ETH, 85%).
2. **Baby.** Hatching is free. The baby earns the lowest tier of daily reward.
3. **Teen.** Minted by feeding the NFT $1\times$ the original egg price (OEP). The teen earns a higher daily reward.
4. **Adult.** Minted by feeding $2\times$ OEP. The adult earns the highest daily reward and unlocks breeding.

Adults of the same species can be bred to mint a new egg for $3\times$ OEP, which goes to the DAO rather than as staked collateral.

2 The pre-drop whitelist

Before each drop, users on the whitelist can purchase one random egg — of any rarity tier — for the price of the most common egg. This means a whitelisted user might pay 0.05 ETH and discover they own a Sublime egg worth 1 ETH. This drop mechanic incentivizes early community participation without requiring users to gamble on rarity tiers.

3 The boost ladder

Daily rewards are amplified by a boost multiplier keyed to the dollar value of staked collateral:

- \$10–\$99: $1.1\times$.
- \$100–\$999: $1.2\times$.
- \$1,000–\$9,999: $1.3\times$.
- \$10,000+: $1.4\times$.

The yield formula is $\text{daily reward} + (\text{collateral} \times \text{boost} \times \text{days}/365)$. The Zoo Reward Calculator surfaces projected earnings.

4 The breeding curve

Each egg generation has a one-baby/one-teen/one-adult mint limit. Adult animals with breeding privileges can breed with another adult of the same species to mint the next generation egg. The breeding-allowance curve descends:

- Gen 1: 7 allowances.
- Gen 2: 6.
- Gen 3: 5.
- Gen 4: 4.
- Gen 5: 3.
- Gen 6: 2.
- Gen 7: 0.

Adults can be sold with or without remaining breeding privileges. This curve enforces scarcity.

5 Daily reward schedule by rarity

- **Sublime:** Baby 1 \$ZOO; Teen 2 \$ZOO; Adult 3 \$ZOO.
- **Rare:** Baby 0.25 \$ZOO; Teen 0.5 \$ZOO; Adult 1 \$ZOO.
- **Endangered:** Baby 0.05 \$ZOO; Teen 0.1 \$ZOO; Adult 0.2 \$ZOO.

A Sublime adult earns $60\times$ the daily reward of an Endangered baby. The reward asymmetry reflects the rarity asymmetry.

6 Play-and-earn games

Two flagship mini-games:

Coin Runner. An endless-runner mechanic. Player dashes through a temple collecting gold/silver coins of escalating denomination, ending in a bullion bar. All rewards redeemable for physical precious metals.

Memory Bridge. A pattern-memory mechanic. Player crosses a bridge in 4 seconds per level by stepping on the correct brick. Bridge lengthens as levels progress.

Both games surface the Lux Token (gold/silver-backed) as the rewards substrate, tying gameplay rewards to commodity-backed value.

7 Asset transfer

Two ways to extract NFT value:

1. **Marketplace sale.** Sell at a price exceeding purchase + collateral + accrued rewards. Buyer inherits all collateral, accrued rewards, and breeding privileges.
2. **Burn.** Destroys the NFT and refunds collateral + accrued rewards, minus the 20% sustainability tax (10% DAO, 10% liquidity pool).

Burning ends the lineage of that NFT permanently, removing it from circulation.

8 Why this is durable gameplay

The Zoo gameplay loop is durable because every action surfaces at least one of: economic upside, conservation impact, or emotional bond. There is no purely speculative move. Even burning — which seems extractive — routes 10% to the DAO conservation treasury. Even breeding — which seems like dilution — pays the 3× OEP fee to the DAO.

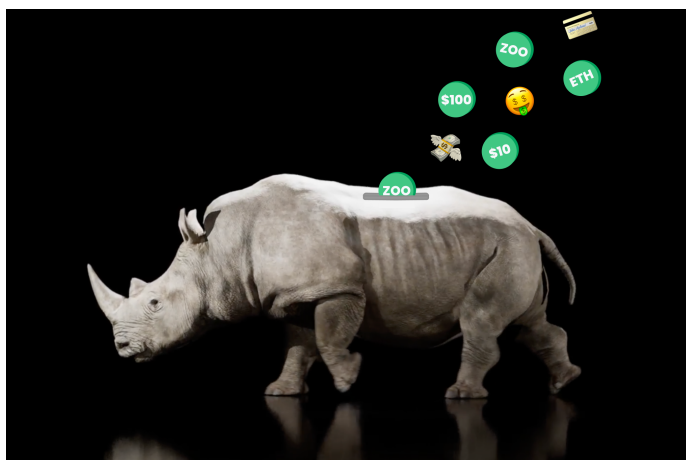


Figure 1: Collateral diagram. Feeding an animal stakes value into its NFT, which then earns boosted daily rewards proportional to the staked amount.

9 Pools

Beyond single-NFT trades, holders can group multiple NFTs into a *pool* listed at a single set price. The buyer pays the pool price (in any cryptocurrency or fiat, converted to \$ZOO under the hood) and atomically receives every NFT in the pool, along with all collateral and accrued rewards. Pools accelerate liquidity for collectors who want to exit a curated collection.

10 Marketplace UX commitments

Per [1], the Zoo Marketplace commits to:

- Crypto and credit/debit-card payment paths in a single seamless checkout.
- Sellers are paid in their listed currency regardless of how the buyer paid.
- Direct \$ZOO purchases via card to lower the on-ramp barrier for new entrants.
- Asset Transfer flow that hands over NFT *plus its stored value* in one transaction.
- Leaderboard surfacing top NFTs and top users by cumulative collection value.

These choices reflect a deliberate stance: Zoo wants to onboard the gaming demographic, not only the existing crypto-native demographic.

11 Marketplace fees

Listing and selling on the Zoo Market is **free**. The fees that apply are:

- **Royalty (2.5%)** — charged on every sale or mint, paid to the original creator.
- **Hatching** — free; effectively burns the egg and mints the baby.
- **Mint Teen** — $1\times$ original egg price (OEP), staked into the NFT.
- **Mint Adult** — $2\times$ OEP, staked.
- **Breeding** — $3\times$ OEP, paid to the DAO (not staked).

12 Sustainability tax breakdown

The whitepaper [1] specifies the following tax schedule:

- **\$ZOO Liquidity Pool Tax (8%)**: 4% to LP, 4% to DAO. Applied on every sell.
- **KEEPER Liquidity Pool Tax (8%)**: 4% to LP, 4% to DAO. Applied on every sell.
- **NFT Burn Tax (20%)**: 10% to LP, 10% to DAO. Applied when collateral-backed NFT is liquidated via burn.

These splits are designed to keep liquidity pools healthy while building the conservation treasury.

References

- [1] Antje Worring. *Zoo: A Game-Fi Multiverse Ecosystem for the Preservation of Endangered Species*. Zoo Labs Foundation, October 31, 2021.